



DEPARTAMENTO DE CIENCIAS DE LA COMPUTACIÓN

CARRERA:

INGENIERÍA DE SOFTWARE

NRC:

2546

ASIGNATURA:

APLICACIONES DISTRIBUIDAS

DOCENTE:

DARIO MORALES

TEMA:

USO DE ORM

NOMBRES:

RAMOS ADRIAN

FECHA:

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1. Introducción

En este informe se detalla el desarrollo de un sistema de gestión de microcursos utilizando tecnologías como Spring Boot, Docker, MySQL y Hibernate. Se presenta una visión general del proyecto, la configuración del entorno, la implementación de las funcionalidades CRUD y el despliegue mediante contenedores Docker.

2. Marco Teórico

Spring Boot

Spring Boot es un framework que facilita la creación de aplicaciones Java al proporcionar configuraciones predeterminadas y herramientas para la integración rápida de tecnologías como bases de datos, seguridad, y servicios REST. Se basa en el ecosistema de Spring Framework y es ampliamente utilizado para aplicaciones modernas.

Docker

Docker es una plataforma de contenedores que permite empaquetar aplicaciones y sus dependencias en un entorno aislado, garantizando consistencia entre entornos de desarrollo y producción.

Hibernate

Hibernate es un framework ORM (Object-Relational Mapping) que simplifica la interacción entre aplicaciones Java y bases de datos relacionales, permitiendo trabajar con objetos en lugar de sentencias SQL.

MySQL

MySQL es un sistema de gestión de bases de datos relacional de código abierto, ampliamente utilizado por su eficiencia, escalabilidad y facilidad de uso.

3. Desarrollo

Configuración del Proyecto

1. Creación del proyecto con Spring Initializr:

- Dependencias: Spring Web, Spring Data JPA, MySQL Driver, Spring Boot DevTools.
- Generación del proyecto base.

2. Estructura del proyecto:

- Paquete model.entity: Definición de las entidades (Course, Student).
- Paquete repositories: Interfaces para operaciones CRUD.
- Paquete services: Lógica de negocio.
- Paquete controllers: Endpoints REST para acceso a la API.

3. Código:

- Entidades (model.entity)

- Curso

```
6  @Entity
7  @Table(name="Cursos")
8  public class Curso {
9
10     @Id
11     @GeneratedValue(strategy = GenerationType.IDENTITY)
12     private long id;
13
14     @Column(nullable = false) 2 usages
15     private String nombre;
16
17     @Column(nullable = false) 2 usages
18     private String descripcion;
19
20     @Column(nullable = false) 2 usages
21     private String credits;
22
23     @Column(name = "fecha_creacion", updatable = false, nullable = false) 3 usages
24     private LocalDateTime fechaCreacion;
25
26     @PrePersist
27     protected void prePersist() {
28         this.fechaCreacion = LocalDateTime.now();
29     }
29 }
```

- Estudiante

```

7  @Entity
8  @Table(name = "Estudiantes")
9  public class Estudiante {
10
11      @Id
12      @GeneratedValue(strategy = GenerationType.IDENTITY)
13      private Long id;
14
15      @Column(nullable = false) 2 usages
16      private String nombre;
17
18      @Column(nullable = false) 2 usages
19      private String apellido;
20
21      @Column(nullable = false, unique = true) 2 usages
22      private String email;
23
24      @Column(name = "fecha_nacimiento", nullable = false) 2 usages
25      @Temporal(TemporalType.DATE)
26      private Date fechaNacimiento;
27
28      @Column(nullable = false) 2 usages
29      private String telefono;
30
31      @Column(name = "fecha_creacion", updatable = false, nullable = false) 3 usages
32      private LocalDateTime fechaCreacion;
33
34      @PrePersist
35      protected void prePersist() {
36          this.fechaCreacion = LocalDateTime.now();
37      }

```

- **Repositories**

- CursoRepository

```

1  package com.espe.micro_cursos.repositories;
2
3  import com.espe.micro_cursos.model.entity.Curso;
4  import org.springframework.data.repository.CrudRepository;
5
6  public interface CursoRepository extends CrudRepository<Curso, Long> { 2 usages
7

```

- EstudianteRepository

```

1  package com.espe.micro_cursos.repositories;
2
3  import com.espe.micro_cursos.model.entity.Estudiante;
4  import org.springframework.data.repository.CrudRepository;
5
6  public interface EstudianteRepository extends CrudRepository<Estudiante, Long> { 2 usages
7
8  }

```

- **Services**

- CursoService

```

1  package com.espe.micro_cursos.services;
2
3  import com.espe.micro_cursos.model.entity.Curso;
4
5  import java.util.List;
6  import java.util.Optional;
7
8  public interface CursoService { 3 usages 1 implementation
9      List<Curso> findAll(); 1 usage 1 implementation
10     Optional<Curso> findById (Long id); 3 usages 1 implementation
11     Curso save(Curso curso); 2 usages 1 implementation
12     void deleteById(Long id); 1 usage 1 implementation
13
14 }

```

- CursoServiceImplement

```
1 package com.espe.micro_cursos.services;
2
3 import com.espe.micro_cursos.model.entity.Cursor;
4 import com.espe.micro_cursos.repositories.CursorRepository;
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.stereotype.Service;
7
8 import java.util.List;
9 import java.util.Optional;
10
11 @Service
12 public class CursoServiceImplement implements CursoService{
13     @Autowired
14     private CursorRepository repository;
15
16     @Override 1 usage
17     public List<Curso> findAll() { return (List<Curso>) repository.findAll(); }
18
19     @Override 3 usages
20     public Optional<Curso> findById(Long id) { return repository.findById(id); }
21
22     @Override 2 usages
23     public Curso save(Curso curso) { return repository.save(curso); }
24
25     @Override 1 usage
26     public void deleteById(Long id) { repository.deleteById(id); }
27 }
```

- EstudianteService

```
1 package com.espe.micro_cursos.services;
2
3 import com.espe.micro_cursos.model.entity.Estudiante;
4
5 import java.util.List;
6 import java.util.Optional;
7
8 public interface EstudianteService { 3 usages 1 implementation
9     List<Estudiante> findAll(); 1 usage 1 implementation
10     Optional<Estudiante> findById(Long id); 3 usages 1 implementation
11     Estudiante save(Estudiante estudiante); 2 usages 1 implementation
12     void deleteById(Long id); 1 usage 1 implementation
13 }
```

- EstudianteServiceImplement

```
4 import com.espe.micro_cursos.repositories.EstudianteRepository;
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.stereotype.Service;
7
8 import java.util.List;
9 import java.util.Optional;
10
11 @Service
12 public class EstudianteServiceImplement implements EstudianteService {
13     @Autowired
14     private EstudianteRepository repository;
15
16     @Override 1 usage
17     public List<Estudiante> findAll() {
18         return (List<Estudiante>) repository.findAll();
19     }
20
21     @Override 3 usages
22     public Optional<Estudiante> findById(Long id) {
23         return repository.findById(id);
24     }
25
26     @Override 2 usages
27     public Estudiante save(Estudiante estudiante) {
28         return repository.save(estudiante);
29     }
30
31     @Override 1 usage
32     public void deleteById(Long id) {
33         repository.deleteById(id);
34     }
35 }
```

- Controladores
 - CursoController

```
13 @RestController
14 @RequestMapping("/api/cursos")
15 public class CursoController {
16     @Autowired
17     private CursoService service;
18     // Crear un curso
19     @PostMapping
20     public ResponseEntity<> crear(@RequestBody Curso curso) {
21         // La fecha de creación se asignará automáticamente en el método @PrePersist
22         Curso nuevoCurso = service.save(curso);
23         return ResponseEntity.status(HttpStatus.CREATED).body(nuevoCurso);
24     }
25     // Obtener todos los cursos
26     @GetMapping
27     public List<Curso> listar() {
28         return service.findAll();
29     }
30     // Obtener un curso por ID
31     @GetMapping("/{id}")
32     public ResponseEntity<> buscarPorId(@PathVariable Long id) {
33         Optional<Curso> curso = service.findById(id);
34         if (curso.isPresent()) {
35             return ResponseEntity.ok(curso.get());
36         }
37         return ResponseEntity.status(HttpStatus.NOT_FOUND).body("Curso no encontrado");
38     }
39     // Actualizar un curso
40     @PutMapping("/{id}")
41     public ResponseEntity<> actualizar(@RequestBody Curso curso, @PathVariable Long id) {
42         Optional<Curso> cursoExistente = service.findById(id);
43         if (cursoExistente.isPresent()) {
44             Curso cursoActualizado = cursoExistente.get();
45             cursoActualizado.setNombre(curso.getNombre());
46             cursoActualizado.setDescripcion(curso.getDescripcion());
47             cursoActualizado.setCreditos(curso.getCreditos());
48             // La fecha de creación no se modifica porque está configurada como "updatable = false"
49             return ResponseEntity.ok(service.save(cursoActualizado));
50         }
51         return ResponseEntity.status(HttpStatus.NOT_FOUND).body("Curso no encontrado");
52     }
53     // Eliminar un curso
54     @DeleteMapping("/{id}")
55     public ResponseEntity<> eliminar(@PathVariable Long id) {
56         Optional<Curso> curso = service.findById(id);
57         if (curso.isPresent()) {
58             service.deleteById(id);
59             return ResponseEntity.ok("Curso eliminado correctamente");
60         }
61         return ResponseEntity.status(HttpStatus.NOT_FOUND).body("Curso no encontrado");
62     }
63 }
```

- EstudianteController

```
13 @RestController
14 @RequestMapping("/api/estudiantes")
15 public class EstudianteController {
16     @Autowired
17     private EstudianteService service;
18     // Crear un estudiante
19     @PostMapping
20     public ResponseEntity<> crear(@RequestBody Estudiante estudiante) {
21         return ResponseEntity.status(HttpStatus.CREATED).body(service.save(estudiante));
22     }
23     // Obtener todos los estudiantes
24     @GetMapping
25     public List<Estudiante> listar() {
26         return service.findAll();
27     }
28     // Obtener un estudiante por ID
29     @GetMapping("/{id}")
30     public ResponseEntity<> buscarPorId(@PathVariable Long id) {
31         Optional<Estudiante> estudiante = service.findById(id);
32         if (estudiante.isPresent()) {
33             return ResponseEntity.ok(estudiante.get());
34         }
35         return ResponseEntity.status(HttpStatus.NOT_FOUND).body("Estudiante no encontrado");
36     }
37 }
```

```

42 // Actualizar un estudiante
43 @PutMapping("/{id}")
44 public ResponseEntity<?> actualizar(@RequestBody Estudiante estudiante, @PathVariable Long id) {
45     Optional<Estudiante> estudianteExistente = service.findById(id);
46     if (estudianteExistente.isPresent()) {
47         Estudiante estudianteActualizado = estudianteExistente.get();
48         estudianteActualizado.setNombre(estudiante.getNombre());
49         estudianteActualizado.setApellido(estudiante.getApellido());
50         estudianteActualizado.setEmail(estudiante.getEmail());
51         estudianteActualizado.setFechaNacimiento(estudiante.getFechaNacimiento());
52         estudianteActualizado.setTelefono(estudiante.getTelefono());
53         return ResponseEntity.ok(service.save(estudianteActualizado));
54     }
55     return ResponseEntity.status(HttpStatus.NOT_FOUND).body("Estudiante no encontrado");
56 }
57
58 // Eliminar un estudiante
59 @DeleteMapping("/{id}")
60 public ResponseEntity<?> eliminar(@PathVariable Long id) {
61     Optional<Estudiante> estudiante = service.findById(id);
62     if (estudiante.isPresent()) {
63         service.deleteById(id);
64         return ResponseEntity.ok( body: "Estudiante eliminado correctamente");
65     }
66     return ResponseEntity.status(HttpStatus.NOT_FOUND).body("Estudiante no encontrado");
67 }
68 }

```

- Configuración de Docker
 - docker-compose.yml

```

1  version: '3.8'
2  services:
3    mysql-db:
4      image: mysql:8.0
5      container_name: mysql-container
6      environment:
7        MYSQL_ROOT_PASSWORD: root
8        MYSQL_DATABASE: sisdb2024
9      ports:
10       - "3307:3306"
11      volumes:
12       - db_data:/var/lib/mysql
13    micro-cursos:
14      build:
15        context: .
16      container_name: micro-cursos-container
17      ports:
18       - "8002:8002"
19      environment:
20        SPRING_DATASOURCE_URL: jdbc:mysql://mysql-db:3306/sisdb2024
21        SPRING_DATASOURCE_USERNAME: root
22        SPRING_DATASOURCE_PASSWORD: root
23      depends_on:
24       - mysql-db
25      volumes:
26       db_data:

```

- DOCKERFILE

```

1  # Usar una imagen base de OpenJDK
2  FROM openjdk:17-jdk-slim
3
4  # Establecer el directorio de trabajo dentro del contenedor
5  WORKDIR /app
6
7  # Copiar el archivo JAR de la aplicación al contenedor
8  COPY target/micro-cursos-0.0.1-SNAPSHOT.jar /app/micro-cursos.jar
9
10 # Exponer el puerto que usará la aplicación
11 EXPOSE 8002
12
13 # Ejecutar el archivo JAR cuando el contenedor arranque
14 ENTRYPOINT ["java", "-jar", "micro-cursos.jar"]

```

Comandos de Construcción y Despliegue

- **Construir el proyecto:**

```
mvn clean package
```

```

PS D:\SEMESTRE OCT24-MAR25\DISTRIBUIDAS\PARCIAL2\CLASE\micro-cursos> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] ----- com.espe:micro-cursos -----
[INFO] Building micro-cursos 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] ----- [ jar ] -----
[INFO]
[INFO] --- clean:3.4.0:clean (default-clean) @ micro-cursos ---
[INFO] Deleting D:\SEMESTRE OCT24-MAR25\DISTRIBUIDAS\PARCIAL2\CLASE\micro-cursos\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ micro-cursos ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ micro-cursos ---
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-utils/3.4.2/maven-sh
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-utils/3.4.2/maven-sha
red-utils-3.4.2.jar (151 kB at 175 kB/s)
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-incremental/1.1/mave
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-incremental/1.1/mave
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-java/1.2.0/plexus-java-1.2.0.j
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-java/1.2.0/plexus-java-1.2.0.j
ar
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/com/thoughtworks/dox/dox2/0.3/dox2-0.3.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/2.15.0/plexus-com
piler-api-2.15.0.jar
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-manager/2.15.0/plexu
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-manager/2.15.0/plexu
s-compiler-manager-2.15.0.jar
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/2.15.0/plexus-
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/2.15.0/plexus-
dependency-tree-3.2.1.jar (43 kB at 11 kB/s)
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.13.0/commons-io-2.13.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aether-api/1.0.0.v20140518/aether-api-1
.0.0.v20140518.jar (186 kB at 31 kB/s)
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/vafer/jdependency/2.8.0/jdependency-2.8.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aether-util/1.0.0.v20140518/aether-util
-1.0.0.v20140518.jar (146 kB at 33 kB/s)
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-collections4/4.4/commons-colle
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-collections4/4.4/commons-colle
ctions4-4.4.jar (484 kB at 106 kB/s)
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/vafer/jdependency/2.8.0/jdependency-2.8.0.jar (233 kB
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/vafer/jdependency/2.8.0/jdependency-2.8.0.jar (233 kB
at 106 kB/s)
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/jdom/jdom2/2.0.6.1/jdom2-2.0.6.1.jar (328 kB at 67 kB
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/jdom/jdom2/2.0.6.1/jdom2-2.0.6.1.jar (328 kB at 67 kB
s)
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/net/java/dev/jna/jna-platform/5.13.0/jna-platform-5.13.0.j
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/net/java/dev/jna/jna-platform/5.13.0/jna-platform-5.13.0.j
na-platform-5.13.0.jar (752 kB at 123 kB/s)
[INFO]
[INFO] [INFO] Replacing main artifact D:\SEMESTRE OCT24-MAR25\DISTRIBUIDAS\PARCIAL2\CLASE\micro-cursos\target\micro-cursos-0.0
[INFO] 1-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/
[INFO] The original artifact has been renamed to D:\SEMESTRE OCT24-MAR25\DISTRIBUIDAS\PARCIAL2\CLASE\micro-cursos\target\
micro-cursos-0.0.1-SNAPSHOT.jar.original
[INFO]
[INFO] ----- BUILD SUCCESS -----
[INFO]
[INFO] Total time: 32.945 s
[INFO] Finished at: 2025-01-08T14:59:08-05:00

```

- **Construir contenedores:**

docker-compose build

```
PS D:\SEMESTRE OCT24-MAR25\DISTRIBUIDOS\PARCIAL2>CLASE\micro-cursos\ docker-compose build
time="2025-01-08T15:00:18-05:00" level=warning msg="D:\SEMESTRE OCT24-MAR25\DISTRIBUIDOS\PARCIAL2\CLASE\micro-cursos\docker-compose.yml: the attribute 'build' is obsolete, it will be ignored, please remove it to avoid potential confusion"

[*] Building 156.7% (18/10) FINISHED
[+] Building 156.7% (18/10) FINISHED docker:desktop-linux
--> transferring dockerfile: 473B 0.0s
--> [spring-boot-app internal] load build definition from Dockerfile 0.0s
--> [spring-boot-app internal] load metadata for docker.io/library/openjdk:17-jdk-slim 3.2s
--> [spring-boot-app auth] library/openjdk:pull token for registry-1.docker.io 0.0s
--> [spring-boot-app internal] load .dockerignore 0.0s
--> transferring context: 2B 0.0s
--> [spring-boot-app 1/3] FROM docker.io/library/openjdk:17-jdk-slim@sha256:aaa33cb27e3e520bb8f116863d0580c438e 152.7s
--> resolve docker.io/library/openjdk:17-jdk-slim@sha256:aaa33cb27e3e520bb8f116863d0580c438e55efca0b126b41f6 0.0s
--> sha256:0d3aa8087687049d0d51808beced9ae21f8f421f84a43263cf8a59e1d6 1.58MB / 1.58MB 0.0s
--> sha256:0d3aa8087687049d0d51808beced9ae21f8f421f84a43263cf8a59e1d6 157.98MB / 157.98MB 0.0s
--> sha256:aaa33cb27e3e520bb8f116863d0580c438e55efca0b126b41f68c362f9774 547B / 547B 0.0s
--> sha256:779635c8cd323c8dbab2d8c4ecf2a920198dfc84bc279824b0d58d22 953B / 953B 0.0s
--> sha256:37cb4321d8023c5f766a3bff65da498478e42cd338091f78531053425d 4.80kB / 4.80kB 0.0s
--> sha256:1a1272f2b2c19f26c650d901aee1a2363 31.30MB / 31.30MB 0.0s
--> extracting sha256:1af617e285f9a6ade12071142c11b7711bc1162c42a586dbd094a1ee2 2.5s
--> extracting sha256:44d3aa807667549d85180bbec9d84e21f0ef44c54b5874422b9c374591d8 0.2s
--> extracting sha256:6c99df1c686bd02f6ad68b133487852854df548b7846c6e88a7f32de 3.1s
--> [spring-boot-app internal] load build context 1.1s
--> transferring context: 54.31MB 1.1s
--> [spring-boot-app 2/3] WORKDIR /app 0.3s
--> [spring-boot-app 3/3] COPY target/micro-cursos-0.0.1-SNAPSHOT.jar /app/micro-cursos.jar 0.1s
--> [spring-boot-app app] exposing to image 0.3s
--> exposing layers 0.2s
--> writing image sha256:e9a6ae6dd2fd2d04992c2839uf6e29dc897b0ddfced32c75727ca673ac712 0.0s
--> naming to docker.io/library/micro-cursos-spring-boot-app 0.0s
```

- **Levantar servicios:**

```
docker-compose up
```

```

D:\SEMESTRE OCT24-MAR25\IDISTRIBUIDAS\PARCIAL2\CLASE[micro-cursos]: docker-compose up --build
time="2025-01-08T20:23:33-05:00" level=warning msg="D:\\SEMESTRE OCT24-MAR25\\IDISTRIBUIDAS\\PARCIAL2\\CLASE[micro-cursos]:docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[*] Building 1.5a (10/10) FINISHED
docker:desktop-linux
=> [micro-cursos internal] load build definition from Dockerfile
0.0s
=> transferring dockerfile: 473B
0.0s
=> [micro-cursos internal] load metadata for docker.io/library/openjdk:17-jdk-slim
1.3s
=> [micro-cursos auth] library/openjdk:pull token for registry-1.docker.io
0.0s
=> [micro-cursos internal] load dockerignore
0.0s
=> transferring context: 2B
0.0s
=> [micro-cursos 1/3] FROM docker.io/library/openjdk:17-jdk-slim@sha256:aea3b3cb27e3e528b8f1a863d8580c438ed55ecfa8bc126b41fe8c3f62f9774
0.0s
=> [micro-cursos internal] load build context
0.0s
=> transferring context: 88B
0.0s
=> CACHED [micro-cursos 2/3] WORKDIR /app
0.0s
=> CACHED [micro-cursos 3/3] COPY target/micro-cursos-0.0.1-SNAPSHOT.jar /app/micro-cursos.jar
0.0s
=> [micro-cursos] exporting to image
0.0s
=> exporting layers
0.0s
=> writing image sha256:58640da1b5005a216a0b1ecd6bbcf8f949364a3731fb4a258ac644571d899
0.0s
=> naming to docker.io/library/micro-cursos:1.5a-micro-cursos
0.0s
=> [micro-cursos] resolving provenance for metadata file
0.0s
[*] Running 1/2
Network micro-cursos_1_default Created 8.1s
Container mysql-container Created 8.1s
Container micro-cursos-container Created 0.1s
Attaching to micro-cursos-container, mysql-container

```

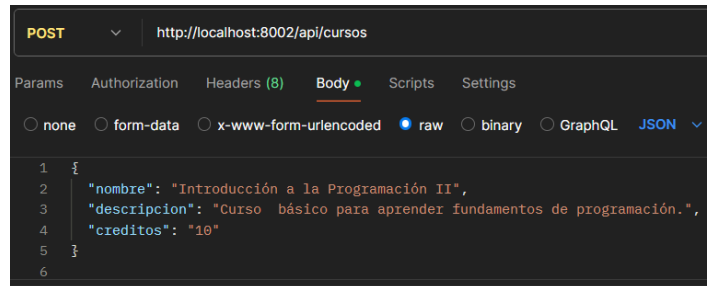
4. Pruebas

Las pruebas del sistema se realizaron utilizando Postman para verificar los endpoints CRUD. A continuación, se listan los pasos y capturas:

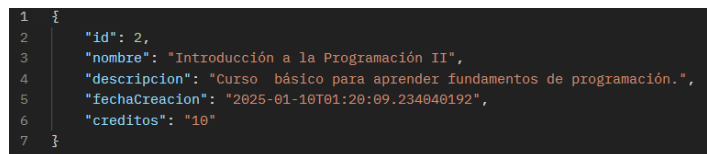
- **Cursos**

1. Crear Curso:

- Endpoint: POST /api/cursos
- Body:

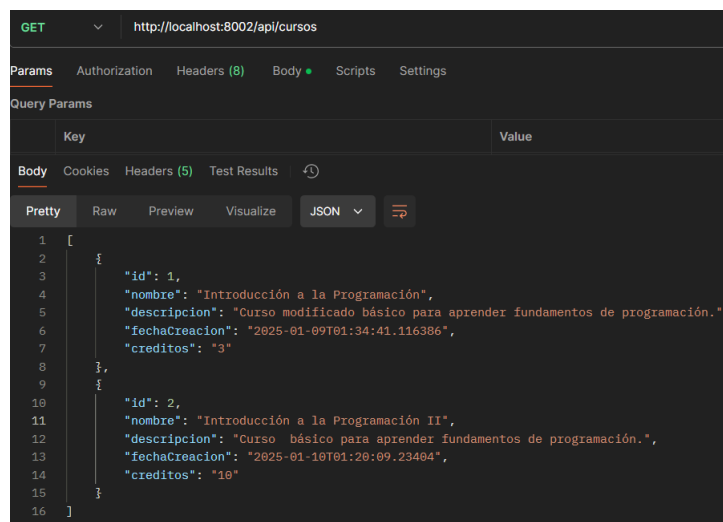


- Resultado:



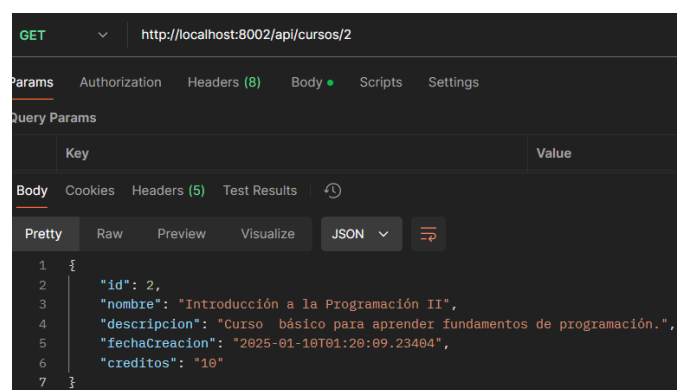
2. Listar Cursos:

- Endpoint: GET /api/cursos



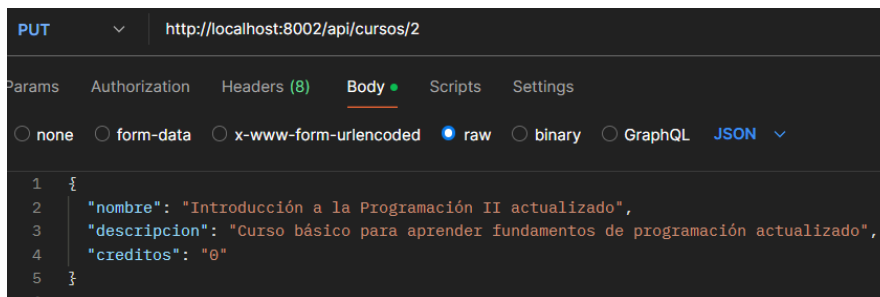
3. Listar Cursos por ID:

- Endpoint: GET /api/cursos/{id}

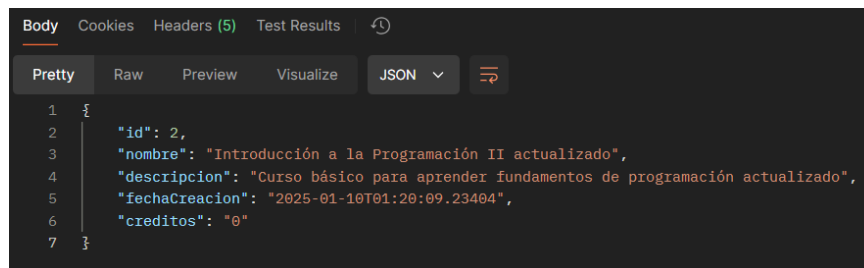


4. Actualizar Curso por ID:

- Endpoint: PUT /api/cursos/{id}
- Body:

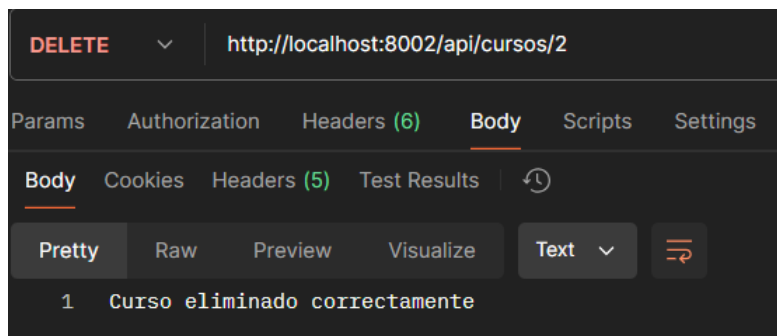


- Resultado:



5. Eliminar Curso por ID:

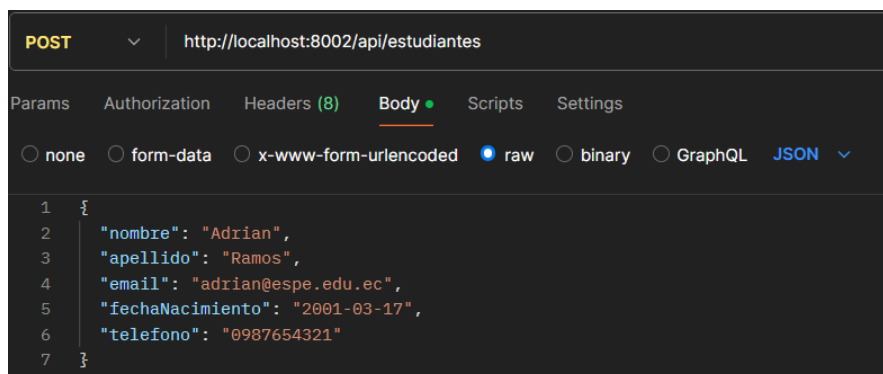
- Endpoint: DELETE /api/cursos/{id}



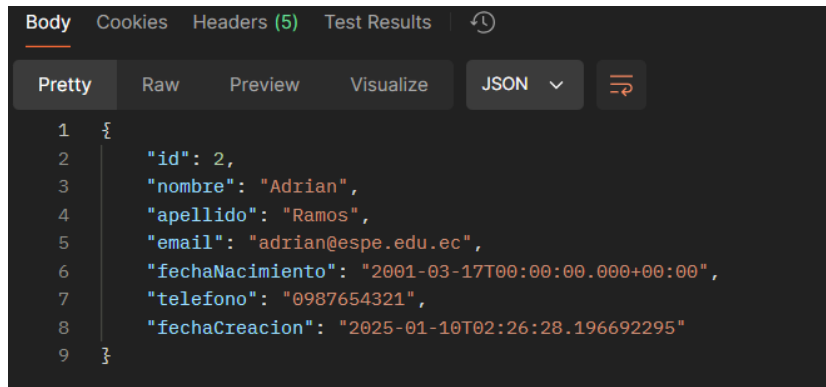
- Estudiantes

1. Crear Curso:

- Endpoint: POST /api/estudiantes
- Body:



- Resultado:

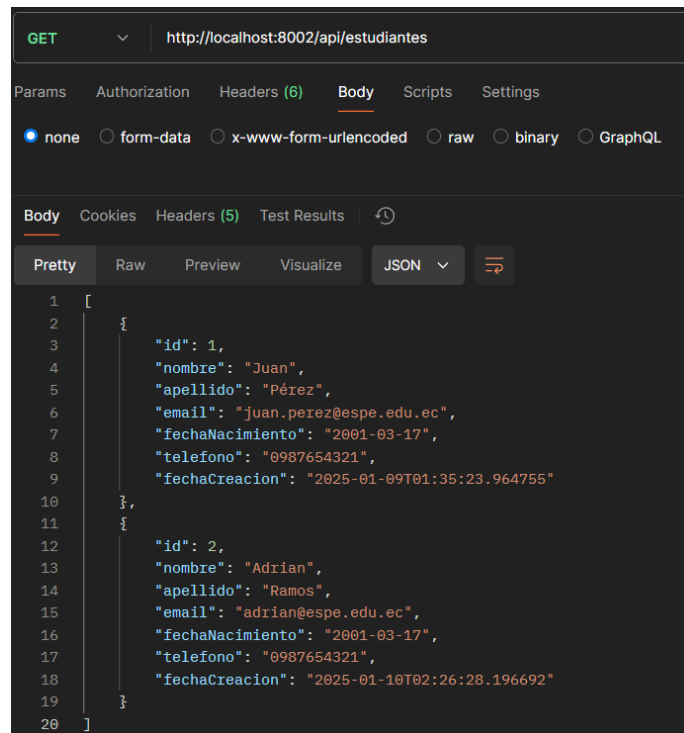


A screenshot of a REST client interface. The top bar shows tabs for 'Body', 'Cookies', 'Headers (5)', and 'Test Results'. The 'Body' tab is active, and the 'Pretty' view is selected. The response is a single JSON object with the following fields: 'id' (2), 'nombre' ('Adrian'), 'apellido' ('Ramos'), 'email' ('adrian@espe.edu.ec'), 'fechaNacimiento' ('2001-03-17T00:00:00.000+00:00'), 'telefono' ('0987654321'), and 'fechaCreacion' ('2025-01-10T02:26:28.196692295').

```
1 {
2   "id": 2,
3   "nombre": "Adrian",
4   "apellido": "Ramos",
5   "email": "adrian@espe.edu.ec",
6   "fechaNacimiento": "2001-03-17T00:00:00.000+00:00",
7   "telefono": "0987654321",
8   "fechaCreacion": "2025-01-10T02:26:28.196692295"
9 }
```

2. Listar Cursos:

- Endpoint: GET /api/estudiantes

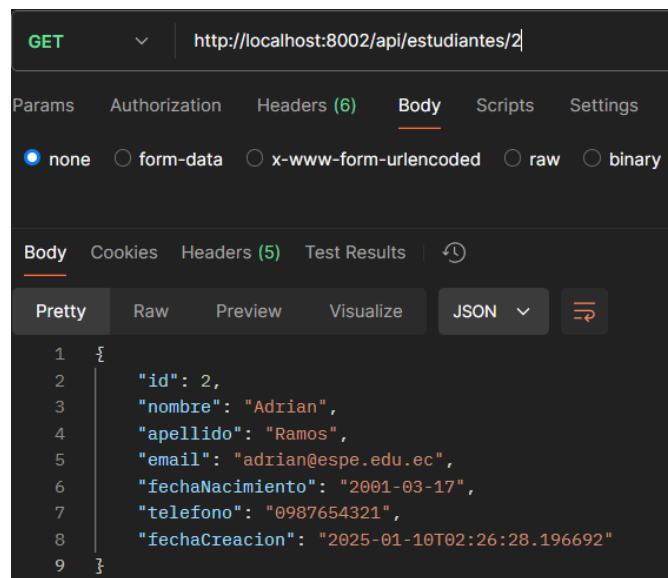


A screenshot of a REST client interface. The top bar shows the method 'GET' and the URL 'http://localhost:8002/api/estudiantes'. The 'Body' tab is active, and the 'Pretty' view is selected. The response is a JSON array containing two objects, each with the same fields as the previous response: 'id', 'nombre', 'apellido', 'email', 'fechaNacimiento', 'telefono', and 'fechaCreacion'.

```
1 [
2   {
3     "id": 1,
4     "nombre": "Juan",
5     "apellido": "Pérez",
6     "email": "juan.perez@espe.edu.ec",
7     "fechaNacimiento": "2001-03-17",
8     "telefono": "0987654321",
9     "fechaCreacion": "2025-01-09T01:35:23.964755"
10  },
11  {
12    "id": 2,
13    "nombre": "Adrian",
14    "apellido": "Ramos",
15    "email": "adrian@espe.edu.ec",
16    "fechaNacimiento": "2001-03-17",
17    "telefono": "0987654321",
18    "fechaCreacion": "2025-01-10T02:26:28.196692"
19  }
20 ]
```

3. Listar Cursos por ID:

- Endpoint: GET /api/estudiantes/{id}

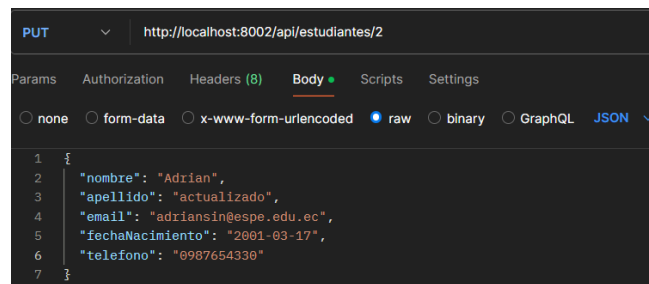


A screenshot of a REST client interface. The top bar shows the method 'GET' and the URL 'http://localhost:8002/api/estudiantes/2'. The 'Body' tab is active, and the 'Pretty' view is selected. The response is a single JSON object for the student with ID 2, containing the same fields as the previous responses.

```
1 {
2   "id": 2,
3   "nombre": "Adrian",
4   "apellido": "Ramos",
5   "email": "adrian@espe.edu.ec",
6   "fechaNacimiento": "2001-03-17",
7   "telefono": "0987654321",
8   "fechaCreacion": "2025-01-10T02:26:28.196692"
9 }
```

4. Actualizar Curso por ID:

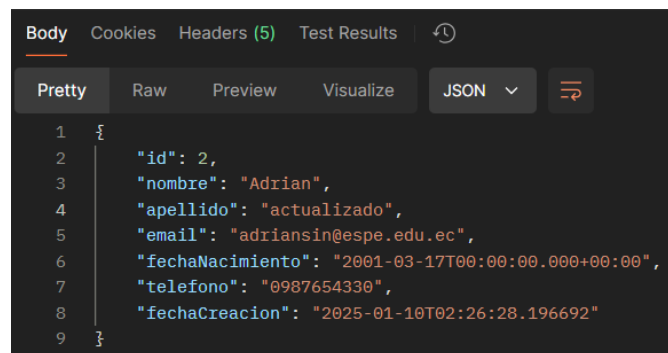
- Endpoint: PUT /api/estudiantes/{id}
- Body:



```
PUT http://localhost:8002/api/estudiantes/2

{
  "nombre": "Adrian",
  "apellido": "actualizado",
  "email": "adriansin@espe.edu.ec",
  "fechaNacimiento": "2001-03-17",
  "telefono": "0987654330"
}
```

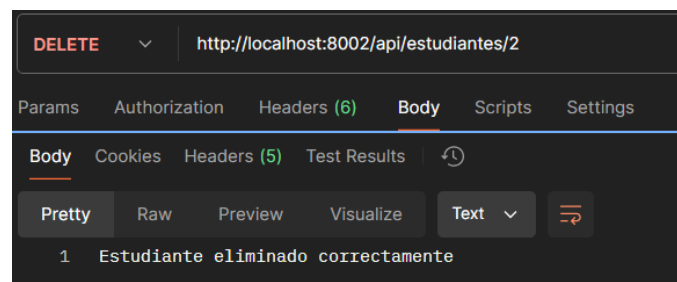
- Resultado:



```
{
  "id": 2,
  "nombre": "Adrian",
  "apellido": "actualizado",
  "email": "adriansin@espe.edu.ec",
  "fechaNacimiento": "2001-03-17T00:00:00.000+00:00",
  "telefono": "0987654330",
  "fechaCreacion": "2025-01-10T02:26:28.196692"
}
```

5. Eliminar Curso por ID:

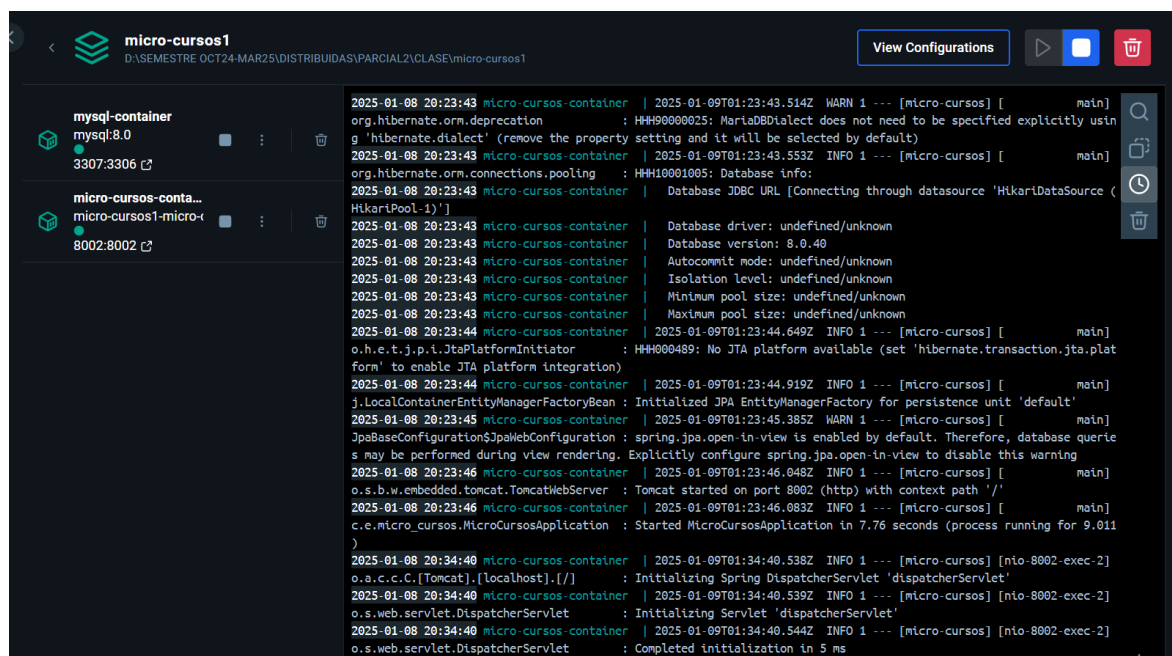
- Endpoint: DELETE /api/estudiantes/{id}



```
DELETE http://localhost:8002/api/estudiantes/2

Estudiante eliminado correctamente
```

- Contenedores levantados



```
mysql-container
mysql:8.0
3307:3306

micro-cursos-conta...
micro-cursos1-micro-t
8002:8002

2025-01-08 20:23:43 micro-cursos-container | 2025-01-09T01:23:43.514Z WARN 1 --- [micro-cursos] [
main] org.hibernate.orm.deprecation : HHH90000025: MariaDBDialect does not need to be specified explicitly usin
g 'hibernate.dialect' (remove the property setting and it will be selected by default)
2025-01-08 20:23:43 micro-cursos-container | 2025-01-09T01:23:43.553Z INFO 1 --- [micro-cursos] [
main] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
2025-01-08 20:23:43 micro-cursos-container | 2025-01-09T01:23:44.649Z INFO 1 --- [micro-cursos] [
main] Database JDBC URL [Connecting through datasource 'HikariDataSource (
HikariPool-1)']
2025-01-08 20:23:43 micro-cursos-container | Database driver: undefined/unknown
2025-01-08 20:23:43 micro-cursos-container | Database version: 8.0.40
2025-01-08 20:23:43 micro-cursos-container | Autocommit mode: undefined/unknown
2025-01-08 20:23:43 micro-cursos-container | Isolation level: undefined/unknown
2025-01-08 20:23:43 micro-cursos-container | Minimum pool size: undefined/unknown
2025-01-08 20:23:43 micro-cursos-container | Maximum pool size: undefined/unknown
2025-01-08 20:23:44 micro-cursos-container | 2025-01-09T01:23:44.649Z INFO 1 --- [micro-cursos] [
main] o.h.e.t.j.p.i.JtaPlatformInitiator : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.plat
form' to enable JTA platform integration)
2025-01-08 20:23:44 micro-cursos-container | 2025-01-09T01:23:44.919Z INFO 1 --- [micro-cursos] [
main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2025-01-08 20:23:45 micro-cursos-container | 2025-01-09T01:23:45.385Z WARN 1 --- [micro-cursos] [
main] JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database querie
s may be performed during view rendering. Explicitly configure spring.jpa.open-in-view to disable this warning
2025-01-08 20:23:46 micro-cursos-container | 2025-01-09T01:23:46.048Z INFO 1 --- [micro-cursos] [
main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8002 (http) with context path '/'
2025-01-08 20:23:46 micro-cursos-container | 2025-01-09T01:23:46.083Z INFO 1 --- [micro-cursos] [
main] c.e.micro_cursos.MicroCursosApplication : Started MicroCursosApplication in 7.76 seconds (process running for 9.011
)
2025-01-08 20:34:40 micro-cursos-container | 2025-01-09T01:34:40.538Z INFO 1 --- [micro-cursos] [nio-8002-exec-2]
o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2025-01-08 20:34:40 micro-cursos-container | 2025-01-09T01:34:40.539Z INFO 1 --- [micro-cursos] [nio-8002-exec-2]
o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2025-01-08 20:34:40 micro-cursos-container | 2025-01-09T01:34:40.544Z INFO 1 --- [micro-cursos] [nio-8002-exec-2]
o.s.web.servlet.DispatcherServlet : Completed initialization in 5 ms
```

5. Conclusión

El proyecto logró implementar un sistema funcional de gestión de microcursos con capacidades CRUD. La integración de Docker permitió un despliegue eficiente y reproducible. Las pruebas confirmaron la funcionalidad y confiabilidad del sistema.

6. Bibliografía

- Walls, C. (2019). Spring Boot in Action. Manning Publications.
- Turnbull, J. (2014). The Docker Book: Containerization is the New Virtualization.
- MySQL Documentation: <https://dev.mysql.com/doc/>
- Docker Documentation: <https://docs.docker.com/>

Link del Github:

https://github.com/adrianmanu/ORM_microcursos